

Assignment – Week 2

Introduction to Machine Learning-IIT KGP

Topics: Linear Regression and Decision Tree

TYPE OF QUESTION: MCQ/MSQ

Number of Questions: 15

Total Marks: $15 \times 2 = 30$

Question 1:

What does entropy measure in the context of a decision tree?

- A. The depth of a decision tree
- B. The number of training samples
- C. The impurity or uncertainty in a dataset
- D. The accuracy of the model on test data

Correct Answer: C. The impurity or uncertainty in a dataset

Detailed Solution : Entropy is a measure of impurity or disorder in a set of examples. A dataset where all examples belong to one class has entropy = 0 (pure). A dataset with equal distribution across classes has maximum entropy. Decision tree algorithms use entropy to decide which attribute to split on.

Question 2:

Suppose in a binary classification dataset, 18 samples belong to class A and 6 samples belong to class B. What is the entropy of the dataset?

- A. 0
- B. 0.81
- C. 1
- D. 1.5

Correct Answer: B

Detailed Solution:

Total samples: $18+6=24$

Probabilities: $p(A)=18/24=0.75$, $p(B)=6/24=0.25$

Entropy= $(-0.75*\log_2(0.75)-0.25*\log_2(0.25)) \approx 0.81$

Question 3:

Information Gain for a split is defined as:

- A. Entropy of child nodes minus entropy of parent node
- B. Sum of entropies of all child nodes
- C. Product of entropy of parent and number of child nodes
- D. Entropy of parent node minus the weighted average entropy of child nodes

Correct Answer: D. Entropy of parent node minus the weighted average entropy of child nodes

Detailed Solution :

Information Gain = $\text{Entropy}(S) - \sum (|S_v|/|S|) \times \text{Entropy}(S_v)$, where S is the full dataset and S_v are the subsets after splitting. It measures how much an attribute reduces uncertainty. The attribute with the highest information gain is chosen as the splitting criterion at each node.

Question 4:

Consider the following training dataset: (x is the input and y is the output)

x	y
1	2
2	5
3	7
4	9

Which of the following linear equations best fits the data?

- A. $y = x + 1$
- B. $y = 2x$
- C. $y = 3x$
- D. $y = x + 4$

Correct Answer: B

Detailed Solution:

Compute the loss function J for each option:

Option A: $J = 29/8 = 3.625$

Option B: $J = 3/8 = 0.375$

Option C: $J = 15/8 = 1.875$

Option D: $J = 11/8 = 1.375$

Option B gives the least error.

Question 5:

A root node of a decision tree has 50 instances and an initial entropy of exactly 0.900. It splits a collection of 50 samples into two subsets. The left sub-node receives 30 instances with an internal Entropy of 0.700. The right sub-node receives the remaining 20 instances with an internal Entropy of 0.400. What is the numerical value of the Information Gain achieved by this split?

- A. 0.460
- B. 0.380
- C. 0.320
- D. 0.900

Correct Answer: C. 0.320

Detailed Solution: Information Gain is calculated by subtracting the weighted average of the children's entropies from the parent node's entropy:

$$\text{Gain} = \text{Entropy}(\text{Parent}) - [(N_{\text{Left}} / N) \times \text{Entropy}(\text{Left}) + (N_{\text{Right}} / N) \times \text{Entropy}(\text{Right})]$$
$$\text{Gain} = 0.900 - [(30 / 50) \times 0.700 + (20 / 50) \times 0.400]$$

$$\text{Gain} = 0.900 - [0.6 \times 0.700 + 0.4 \times 0.400] = 0.900 - [0.420 + 0.160]$$

$$\text{Gain} = 0.900 - 0.580 = 0.320.$$

Question 6:

What is true about Gradient Descent in Linear Regression?

- A. It updates parameters to minimize the cost function.
- B. It always increases training error.
- C. It can only be used for classification problems.
- D. It does not require a learning rate.

Correct Answer: A

Detailed Solution: Gradient descent iteratively updates parameters to minimize the cost function.

Question 7:

Which of the following techniques can help reduce overfitting in decision trees?

- A. Increasing tree depth indefinitely
- B. Pruning
- C. Using fewer training samples
- D. Removing validation data

Correct Answer: B

Detailed Solution: Pruning removes unnecessary branches and reduces model complexity, helping prevent overfitting.

Question 8:

Which of the following are correct relationships between bias, variance, and model behaviour? Select ALL that apply.

- A. High variance corresponds to overfitting
- B. High bias corresponds to underfitting
- C. Increasing model complexity always reduces both bias and variance simultaneously
- D. A model with low bias and low variance is the ideal goal

Answer: A, B, D

Detailed Solution: High variance means the model is overly sensitive to training data (overfitting) — A is correct. High bias means the model is too simple to capture patterns (underfitting) — B is correct. The ideal model has low bias AND low variance — D is correct. However, in practice, reducing bias (increasing complexity) typically increases variance, and vice versa. They cannot generally be reduced simultaneously without more data or better algorithms — C is false.

Question 9:

Consider the following training dataset: (Input Attributes: Outlook, Temperature, Windy; Output: Play)

Outlook	Temperature	Windy	Play
Sunny	Hot	No	No
Sunny	Mild	No	Yes
Rainy	Mild	No	Yes
Rainy	Cool	Yes	No
Overcast	Cool	No	Yes
Overcast	Mild	Yes	Yes

Which attribute is most likely to be selected as the root node of the decision tree using Information Gain?

- A. Temperature
- B. Outlook
- C. Windy
- D. Play

Correct Answer: A

Detailed Solution: The attribute Temperature gives the highest information gain among all attributes and is therefore selected as the root node.

Question 10:

Two linear regression models are trained on the same data. Model X has training MSE = 0.05 and test MSE = 3.80. Model Y has training MSE = 1.20 and test MSE = 1.35. Which model should be preferred for deployment?

- A. Model X, because its training error is much lower
- B. Model Y, because the small gap between training and test error indicates better generalization
- C. Model X, because lower training error always means a better model
- D. Both models are equally good

Correct Answer: B. Model Y, because the small gap between training and test error indicates better generalization

Detailed Solution:

Model X's huge gap between training MSE (0.05) and test MSE (3.80) is a strong indicator of overfitting — it has memorized the training data but fails on new examples. Model Y shows a modest, consistent error on both sets, indicating it has genuinely learned the underlying pattern. For deployment, generalization matters more than low training error.

Question 11:

Why does overfitting become more likely when the training dataset is small?

- A. Small datasets have lower entropy
- B. Small datasets always have high bias
- C. With fewer examples, a complex model can memorize them exactly, failing to capture the true underlying pattern
- D. Small datasets require fewer parameters to fit

Correct Answer: C. With fewer examples, a complex model can memorize them exactly, failing to capture the true underlying pattern

Detailed Solution:

When a training set is small, a sufficiently flexible model can find a hypothesis that passes through every training point exactly. This results in zero training error but poor generalization to unseen data. The model has learned the noise and specific quirks of the small sample rather than the true data distribution.

Question 12:

Which of the following measures is commonly used for selecting the best attribute in a decision tree?

- A. Accuracy
- B. Mean Squared Error only
- C. Precision
- D. Information Gain

Correct Answer: D

Detailed Solution: Decision trees commonly use entropy and information gain to select the attribute that best splits the data.

Question 13:

What role does the **learning rate (α)** play when applying Gradient Descent to optimize a Linear Regression cost function?

- A. It specifies the total number of decision tree branches to construct during initialization.
- B. It defines the step size taken in the direction of the negative gradient to update parameters toward the minimum cost.
- C. It alters the target variable from a continuous number to a categorical distribution.
- D. It guarantees that the training error drops to exactly zero after the very first iteration.

Correct Answer: B

Detailed Solution: Gradient descent iteratively adjusts the model's parameters to minimize the global cost function. The learning rate (α) determines how large of a step the algorithm takes during each parameter update along the gradient's slope. If α is too

small, convergence is slow; if it is too large, the algorithm can overshoot the global minimum.

Question 14:

In standard classical Linear Regression, what is the fundamental nature of the target output variable (y), and how does it differ from a standard classification target?

- A. It is always a categorical text string representing discrete group memberships.
- B. It is strictly limited to binary boolean true/false values.
- C. It is a continuous numerical value that can take any real value.
- D. It does not exist because linear regression is an unsupervised learning procedure.

Correct Answer: C

Detailed Solution: Linear Regression is designed specifically for predicting continuous numerical variables (such as prices, temperatures, or trends). This is in contrast to classification models, where the output is categorical, representing discrete class labels or groups.

Question 15:

In a binary classification setting, a leaf node of a decision tree contains 14 samples from Class A and 2 samples from Class B. What is the Gini Impurity of this node?

- A. 0.219
- B. 0.125
- C. 0.500
- D. 0.000

Correct Answer: A

Detailed Solution:

Total samples at the node: $14+2=16$

Probabilities: $p(A)=14/16=0.875$, $p(B)=2/16=0.125$

The formula for Gini Impurity is:

$$\text{Gini}=1-((0.875)^2+(0.125)^2)$$

$$\text{Gini}=1-(0.765625+0.015625)=1-0.78125=0.21875\approx 0.219$$
